

Submission C solution to Problem 7

Abstract

We show that the complex F_w in the question is not always contractible. We give an explicit reducible quasi-reduced word w for which F_w is isomorphic to a two-dimensional cubical complex with nonzero H_2 . In fact, the relevant subcomplex is a cubical cell structure on S^2 .

1 Two elementary facts about matchings

Let $\mathbb{H} = \mathbb{R} \times \mathbb{R}_+$. Given a matching M , let $\Gamma(M)$ be the finite graph obtained from M by declaring the boundary points and the crossings to be vertices. Thus boundary vertices have valence 1, crossing vertices have valence 4, and the edges of $\Gamma(M)$ are the open subarcs between consecutive vertices.

Lemma 1 (No cycles). *For every matching M satisfying the defining conditions in the question, the graph $\Gamma(M)$ is a forest.*

Proof. Suppose $\Gamma(M)$ had a cycle. Since all valence-one vertices lie on $\mathbb{R} \times \{0\}$, every cycle lies in the interior of $\mathbb{H}$. A finite plane graph with a cycle has a bounded complementary face whose boundary contains no boundary vertex on $\mathbb{R} \times \{0\}$. For the corresponding component R of $\mathbb{H} \setminus M$, some component of $M \cap \partial \bar{R}$ therefore has boundary not equal to two points of $\{1, \dots, 2n\} \times \{0\}$. This contradicts the third condition in the definition of matching. $\square$

Lemma 2 (Labels around one component). *Let T be a connected component of $\Gamma(M)$. Let*

$$p_1, p_2, \dots, p_m$$

be the boundary leaves of T , listed in the cyclic order in which they occur around a small regular neighbourhood of T . Then

$$w_{p_i} = w_{p_{i+1}}^{-1}$$

for every i , with indices taken modulo m . In particular, all leaves of T are labelled by one letter ℓ and its inverse, alternating around T .

Proof. By Lemma 1, T is a tree. A sufficiently small regular neighbourhood $N(T) \subset \mathbb{H}$ is therefore a disk. Its frontier in $\mathbb{H}$ consists of arcs joining consecutive leaves p_i, p_{i+1} in the cyclic order around T . Each such frontier arc is a small push-off of a component of $M \cap \partial \bar{R}$ for a complementary region R . By the defining condition on matchings, the endpoints of that component have inverse labels. Hence $w_{p_i} = w_{p_{i+1}}^{-1}$ for all i . $\square$

Corollary 1 (A uniquely occurring letter is forced). *Suppose a letter x and its inverse x^{-1} occur exactly once each in w . Then in every matching M , the two corresponding boundary points are joined by a single interval with no crossings.*

Proof. Let T be the component of $\Gamma(M)$ containing the unique x -labelled leaf. If T had more than two leaves, then that leaf would have two distinct cyclic neighbours around T , and Lemma 2 would force both of them to be labelled x^{-1} , impossible because x^{-1} occurs only once. Thus T has exactly two leaves.

Since T is a tree with m leaves and v four-valent vertices, the degree count gives

$$m + 4v = 2(m + v - 1).$$

For $m = 2$, this yields $v = 0$. Hence T is a single interval with endpoints labelled x and x^{-1} , and it has no crossings. $\square$

2 The counterexample

Let

$$G = \langle a, b, c \rangle$$

be the free group of rank 3, and write $\bar{a} = a^{-1}$, $\bar{b} = b^{-1}$, $\bar{c} = c^{-1}$. Consider the word

$$w = a\bar{a}b\bar{b}a\bar{a}c\bar{c}a\bar{a}.$$

This word is freely reducible to the identity. It is also quasi-reduced: its length-three subwords are

$$a\bar{a}b, \quad \bar{a}b\bar{b}, \quad b\bar{b}a, \quad \bar{b}a\bar{a}, \quad a\bar{a}c, \quad \bar{a}c\bar{c}, \quad c\bar{c}a, \quad \bar{c}a\bar{a},$$

and none has the form $\ell\ell^{-1}\ell$.

By Corollary 1, the unique occurrences of $b, \bar{b}$ are forced to form a single isolated interval, and similarly the unique occurrences of $c, \bar{c}$ are forced to form a single isolated interval. Therefore deleting those two forced intervals gives a cubical isomorphism from F_w to the matching complex for the alternating six-letter word

$$\alpha = a\bar{a}a\bar{a}a\bar{a}.$$

Conversely, any matching for α gives a matching for w by inserting small isolated intervals for $b\bar{b}$ and $c\bar{c}$ in the two gaps. This operation preserves crossings, resolutions, and attaching maps. Hence it suffices to compute the complex $K := F_\alpha$.

3 The complex for $a\bar{a}a\bar{a}a\bar{a}$

Label the six boundary points $1, \dots, 6$, with odd points labelled a and even points labelled $\bar{a}$. We write (ij) for an interval joining i to j , and products such as $(12)(34)(56)$ for the corresponding matching.

Because of Lemma 1, two intervals in a valid matching cross at most once. Thus an endpoint pairing determines the crossing pattern: two intervals cross exactly when their endpoints interleave. There are 15 pairings of the six endpoints. They fall into the following classes:

crossings	pairings
0	$(12)(34)(56), (12)(36)(45), (14)(23)(56), (16)(23)(45), (16)(25)(34)$
1	$(12)(35)(46), (13)(24)(56), (13)(26)(45), (15)(23)(46), (15)(26)(34), (16)(24)(35)$
2	$(13)(25)(46), (14)(26)(35), (15)(24)(36)$
3	$(14)(25)(36).$

The last pairing has three pairwise crossings and hence contains a cycle in the underlying graph, so it is invalid by Lemma 1. The pairings in the first three rows satisfy the side-label condition. For the two-crossing pairings, the sides of a regular neighbourhood connect the cyclically consecutive pairs

$$(1, 2), (2, 3), (3, 4), (4, 5), (5, 6), (6, 1),$$

all of which have inverse labels.

Thus K has five vertices. Name them

$$N = (12)(34)(56), \quad S = (16)(23)(45),$$

and

$$E_1 = (12)(36)(45), \quad E_2 = (14)(23)(56), \quad E_3 = (16)(25)(34).$$

The six edges are precisely the edges joining each of N, S to each E_i . The three square two-cells are:

$$D_{12} = (13)(25)(46), \quad D_{13} = (14)(26)(35), \quad D_{23} = (15)(24)(36),$$

with boundaries

$$\partial D_{12} = N E_1 S E_2 N,$$

$$\partial D_{13} = N E_1 S E_3 N,$$

$$\partial D_{23} = N E_2 S E_3 N.$$

There are no higher-dimensional cells because a valid matching on three intervals has at most two crossings by Lemma 1.

Proposition 1. *The complex K has nonzero second homology.*

Proof. Orient the edges a_i from N to E_i , and orient the edges b_i from E_i to S . Orient the square D_{ij} by the boundary path

$$N \longrightarrow E_i \longrightarrow S \longrightarrow E_j \longrightarrow N.$$

Then

$$\partial D_{ij} = a_i + b_i - b_j - a_j.$$

Therefore

$$z := D_{12} - D_{13} + D_{23}$$

satisfies

$$\partial z = (a_1 + b_1 - b_2 - a_2) - (a_1 + b_1 - b_3 - a_3) + (a_2 + b_2 - b_3 - a_3) = 0.$$

Since K has no 3-cells, this nonzero cellular 2-cycle represents a nonzero class in $H_2(K; \mathbb{Z})$. $\square$

Theorem 1. *The complex F_w is not contractible for*

$$w = a\bar{a} b\bar{b} a\bar{a} c\bar{c} a\bar{a}.$$

Consequently, the answer to the question is negative in general.

Proof. As shown above, $F_w \cong K$. By Proposition 1,

$$H_2(F_w; \mathbb{Z}) \cong H_2(K; \mathbb{Z}) \neq 0.$$

A contractible CW complex has trivial reduced homology in all positive degrees. Hence F_w is not contractible. $\square$